

⇒ Boosting
↑

✓ Bagging
↑

✓ High var, low Bias

+
✓ Random subsets
(Rs + cs)

+
✓ Aggregation

low var, High Bias
(log Regression)

+
Additive / combine
base

⇓

↑
→ $L_2()$

↓
[84] ←
[83]

DT() → 90%
↑↑
72%

⇒ core idea

$$D = \{x, y\}$$

↓

⇒ at stage '0'

$$D_{\text{Train}}(x, y) \longrightarrow M_0$$

$$y_0 = \hat{y}_0(x)$$

$\vdots$

⇓

$$\{x_i, y_i, \text{error}_i\} \longrightarrow \text{Stage II}$$

$$\{x, \text{error}_i\}$$
$$\boxed{M_1}$$

x	y	z	err
x1	145	✓	—
x2	146	✓	—
x5	147	✓	—

N $\longleftrightarrow \{x_i \text{ envoy}\}$

$\downarrow$
 $h_1(x)$
 $\leftarrow$

$$envoy = \text{envoy} - h_1(x)$$

$$e_1 = e - h_1(x) \rightarrow \hat{y}$$

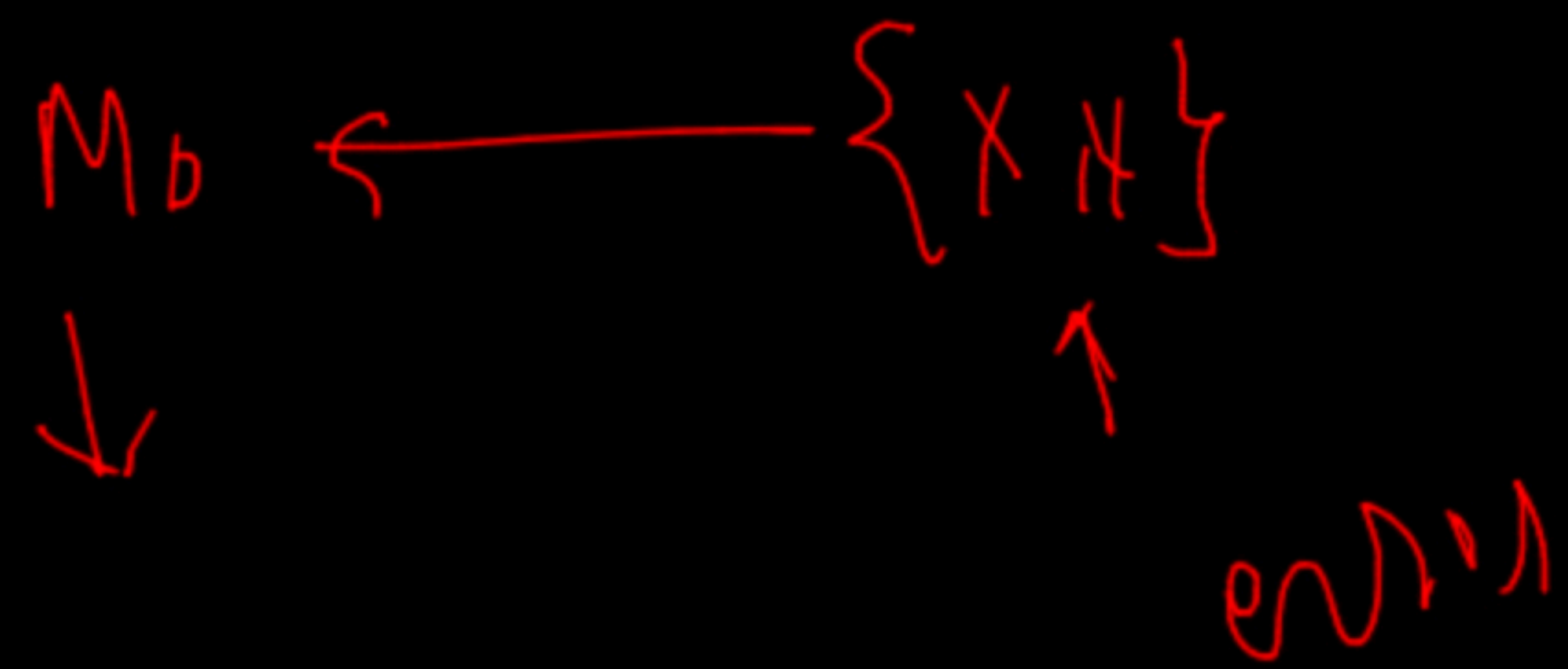
Repeating $\rightarrow 20 \text{ times} \rightarrow \downarrow \downarrow \downarrow$
 $\underline{\underline{=}}$

x	e_0	$\hat{y}$	e_1
x_1	1	✓	
x_2	1	✓	
x_3	1	✓	

$\uparrow$

$\uparrow$

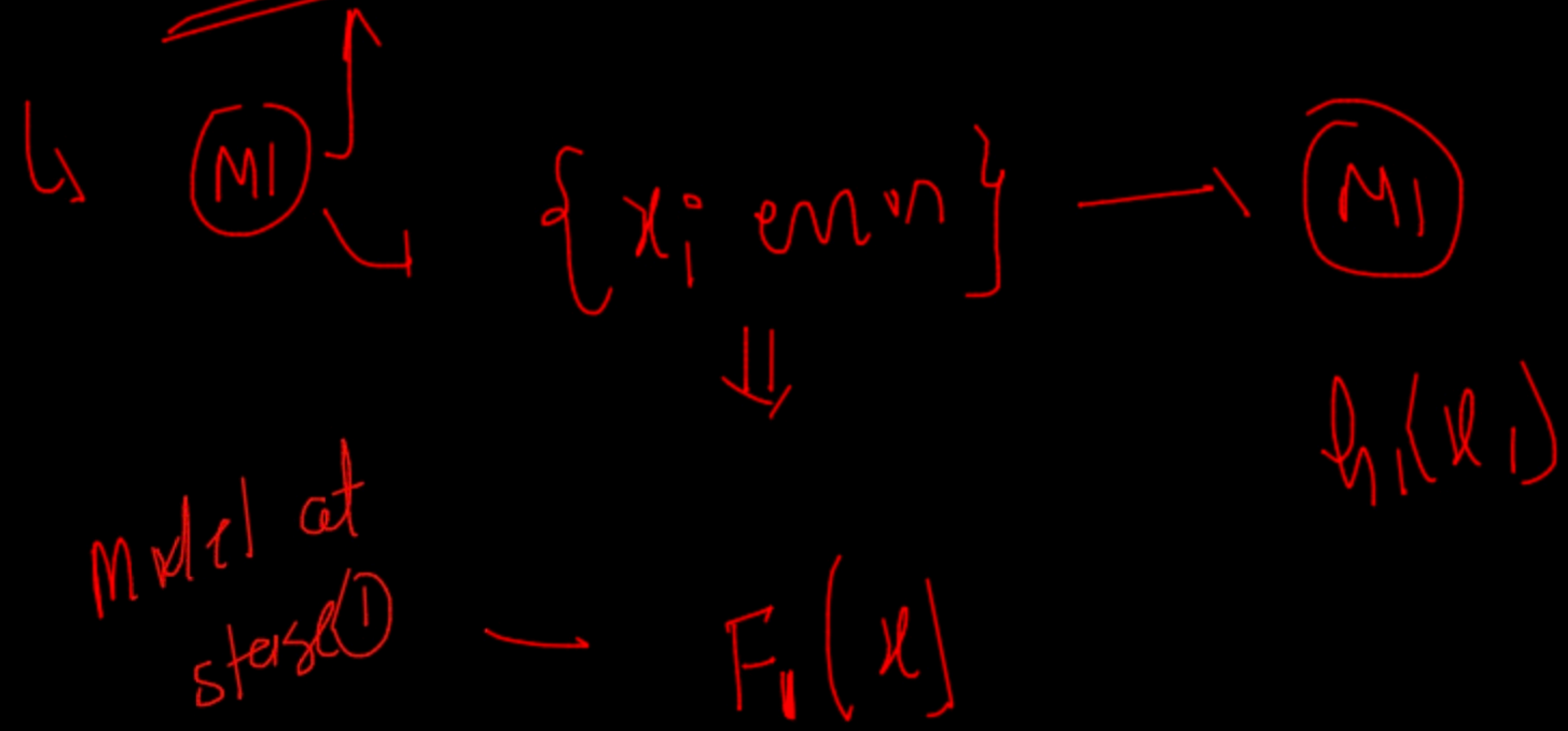
$\downarrow$



801
787-

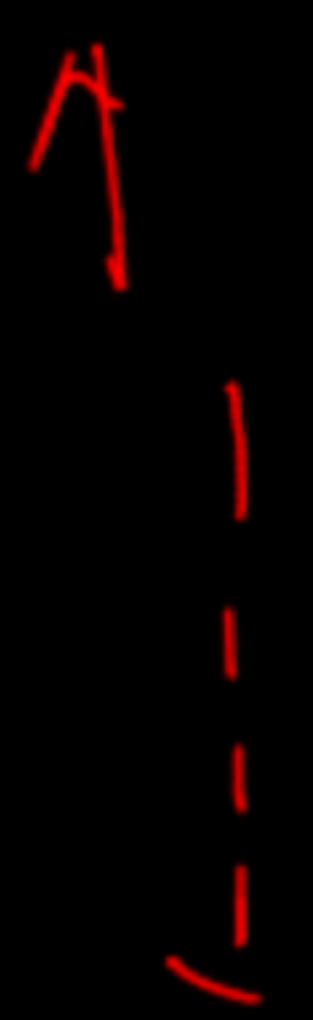


$y = h(x)$
error
 $y_p - h(x)$



Vertical toolbar with icons for various drawing tools: selection, eraser, pencil, highlighter, text, shape, and zoom. At the bottom, it shows '4 / 4'.

$$F_1(x) = \alpha_0 h_0(u) + \alpha_1 h_1(u)$$



kth time →

error 1

error ↓ ↓ ↓

-parameters ↓

24

100

0.2

$$F_k(x) = \sum_{i=1}^k \alpha_i h_i(u)$$



kt = [1, 2, 3, 4, 5]
↑ ↑ 1 1 4

search

for i in kt
if i == 5
→ 1

$O(n^2)$

$O(n^3)$

$O(n)$
↑ ↑

max

sklearn

GBDT

XgBoost

Light GB

or